

AI-Powered Customer Support & Workflow Automation Platform

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PROJECT PROPOSAL: AI-POWERED CUSTOMER SUPPORT & WORKFLOW AUTOMATION PLATFORM

1. PROJECT OVERVIEW

We want to build an AI-driven customer support automation platform that combines a multi-channel inbox (email, live chat, WhatsApp) with LLM-powered ticket triage, automated reply drafting, and no-code workflow automation rules. The goal is to reduce manual support workload for small and mid-sized e-commerce and SaaS companies while keeping a human-in-the-loop for quality control.

2. TARGET USERS

- **Support Agents:** Unified inbox to review AI-drafted replies, escalate complex tickets, and tag conversations.
- **Support Managers:** Dashboard for SLA tracking, automation rule builder, and AI performance analytics.
- **End Customers:** Self-serve chat widget with AI-powered FAQ resolution and ticket creation when human handoff is needed.

3. CORE FEATURES & FUNCTIONAL REQUIREMENTS

- **Multi-Channel Unified Inbox:** Combine email (Gmail API / IMAP), a live chat widget, and WhatsApp Business API into a single ticket queue.
- **LLM-Powered Ticket Triage & Auto-Drafting:** Use an LLM (GPT-4o or Claude) to classify intent, urgency, and sentiment, and auto-draft suggested replies with a human approval step before sending.
- **RAG-Based Knowledge Base Q&A:** Vector-embedded help center content (pgvector or Pinecone) so AI responses stay grounded in company-specific documentation.
- **Workflow Automation Builder:** No-code trigger-condition-action rule builder (similar to Zapier) for auto-tagging, auto-assignment, and auto-escalation.
- **Sentiment & SLA Analytics Dashboard:** Real-time view of response times, CSAT scores, and sentiment trends.
- **Third-Party Integrations:** Slack notifications, Shopify/Stripe lookups for order and billing context, and a public API for custom integrations.
- **Human-in-the-Loop Approval Queue:** Configurable approval gates for AI-drafted responses that fall below a confidence threshold.

4. TECHNICAL SPECIFICATIONS & STACK PREFERENCES

- **Frontend (Web):** Next.js, TypeScript, Tailwind CSS, and TanStack Query for data fetching.
- **Backend:** Node.js (NestJS) with BullMQ for async job queues (embeddings, classification, drafting).
- **Database:** PostgreSQL (tickets, users, rules) and pgvector or Pinecone (embeddings), plus Redis for queues and caching.

- **External APIs:** OpenAI / Anthropic API (LLM tasks), WhatsApp Business API, Gmail API, Shopify Admin API, and Stripe API.
- **Cloud Hosting:** AWS (ECS Fargate, RDS, ElastiCache), with Render or Railway considered for faster initial deployment.

5. ESTIMATION, BUDGET, AND TIMELINE

Estimated Timeline: 3 to 5 months

Budget Range: \$35,000 – \$55,000 USD

Key Milestones:

- **Phase 1: Inbox & Channel Integrations** (Weeks 1-4) – Email, chat widget, and WhatsApp Business API onboarding into a unified ticket queue.
- **Phase 2: LLM Triage & Auto-Drafting Pipeline** (Weeks 5-9) – Intent/urgency/sentiment classification and AI reply drafting with approval workflow.
- **Phase 3: Knowledge Base RAG & Automation Builder** (Weeks 10-14) – Embedding pipeline for help docs and the no-code rule builder.
- **Phase 4: Analytics Dashboard & Integrations** (Weeks 15-18) – SLA/CSAT dashboard, Slack, Shopify, and Stripe integrations.
- **Phase 5: QA, Approval Workflows & Launch** (Weeks 19-20) – End-to-end testing, approval gate tuning, and deployment.